

Optogenetic Sweet Neuron Activation as a Reward for Spatial Place Conditioning and Omission Testing in *Drosophila melanogaster*

Independent Project Proposal – NEU 350

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Introduction

Drosophila melanogaster is a powerful model for studying learning and memory: its compact nervous system, precisely characterized neural circuits, and genetic tractability make it ideal for linking cellular mechanisms to behavior (Tully and Quinn, 1985; Kahsai and Zars, 2011). Associative learning depends critically on the mushroom bodies, where coincidence detection between conditioned and unconditioned stimuli drives synaptic plasticity (Kim et al., 2007). Appetitive reward is conveyed to the mushroom bodies by dopaminergic neurons of the paired anterior medial (PAM) cluster, which respond to sweet taste stimulation (Masek and Scott, 2015; Aso et al., 2014). The Gr64f-Gal4 driver labels peripheral sweet-sensing gustatory receptor neurons (GRNs) that activate PAM neurons and drive both immediate appetitive behavior and associative memory formation (Dahanukar et al., 2007; Scott, 2018).

CsChrimson, a red-light-gated cation channel, expressed in sweet GRNs via Gr64f-Gal4 reliably substitutes for sugar reward: flies strongly prefer locations where CsChrimson activation is triggered, and this effect is ATR-dependent, confirming it requires functional optogenetic stimulation (Yapici et al., 2016; Klapoetke et al., 2014). This approach has been extended to reward-choice paradigms in which Gr64f-driven optogenetic activation reinforces learned preferences, with the key advantage that reward delivery can be precisely tied to spatial location rather than food ingestion (Beron et al., 2023). While *Drosophila* can form spatial preferences using aversive heat-based reinforcement (Wustmann et al., 1996; Foucaud et al., 2010), appetitive spatial conditioning in freely walking flies using a gustatory reward has not been demonstrated (Wiggin et al., 2021).

A further open question is whether any learned spatial preference reflects goal-directed action, sensitive to the action-outcome contingency, or habitual behavior that persists independently of the reward (Dickinson, 1985; Balleine and Dickinson, 1998). The contingency omission test addresses this directly: reward is removed and responding is tracked across extinction trials. Goal-directed animals update their behavior when the contingency breaks down; habitual animals do not (Bouton, 2024). In *Drosophila*, reward omission is processed as a negative prediction error through dopaminergic circuits in the mushroom body, forming a competing extinction memory (Felsenberg et al., 2018), and the mushroom bodies regulate the transition from goal-directed to habitual responding with extended training (Brembs, 2009). We hypothesize that (1) *Drosophila* expressing Gr64f-Gal4 > UAS-CsChrimson will develop a significant arm preference in a T-maze when one arm is paired with red-light stimulation, and (2) this preference will decline across extinction trials when the reward is omitted, consistent with goal-directed rather than habitual responding.

Methods

Fly stocks and preparation. All experiments will use Gr64f-Gal4 x UAS-CsChrimson flies (vial "2-12 / Sweet Response," confirmed available in lab). Flies will be raised on standard cornmeal-dextrose food supplemented with 0.5 mM all-trans-retinal (ATR) in the dark for at least 2 days before experiments; ATR is required for functional CsChrimson expression and was used by instructors for the module flies (Klapoetke et al., 2014; Yapici et al., 2016). ATR availability will be confirmed at the April 1-2 project conference. Control flies are the same genotype raised without ATR. Flies will be food-deprived for 16-20 hours before each session (Masek and Scott, 2015). We will run 3-4 flies per session across 4 sessions (sessions 1-2 for piloting, 3-4 for data collection), targeting $N \geq 10$.

T-maze apparatus. A T-shaped arena with a transparent floor will be fabricated from clear acrylic and sized to fit the window of the existing observation platform. The FLIR Firefly camera (FFY-U3-16S2M) mounted beneath the platform on the Narishige manipulator will capture fly position under infrared illumination. The S-900-controlled red LED above the platform illuminates the full arena by default; the non-rewarded arm and central stem will be masked from above with black cardstock to spatially restrict stimulation to the rewarded arm only. The rewarded arm will be counterbalanced across flies. Stimulus parameters will be optimized during pilot sessions to avoid overwhelming the fly, consistent with module protocol guidance.

Phase 1 – Place conditioning. Each fly will undergo a 5-minute baseline probe (light off), followed by 4-6 training trials of 3 minutes each with red light active in the rewarded arm only. Between trials, the fly will be guided back to the central stem with a brief air puff. A final 5-minute post-training probe (light off) will assess whether a place preference was acquired. Phase 1 constitutes a complete, interpretable experiment independent of Phase 2.

Phase 2 – Omission test. Immediately following Phase 1, flies undergo 3 consecutive 3-minute extinction trials with the LED permanently off in both arms. The fly enters the previously rewarded arm but receives no light. ATR-negative controls undergo the identical sequence. This directly tests whether the learned preference is sensitive to removal of the action-outcome contingency (goal-directed) or persists despite it (habitual) (Dickinson, 1985; Felsenberg et al., 2018).

Behavioral quantification and analysis. Video will be acquired at ~60 fps using SpinView in buffered recording mode (600-frame segments), concatenated across each trial. SLEAP will track head and thorax position using the same pipeline as the module: manual labeling of at least 20 frames per video, single-animal model training, and full-video inference (Pereira et al., 2022). A Preference Index ($PI = [\text{time in rewarded arm} - \text{time in unrewarded arm}] / \text{total arm time}$) will be computed per trial. For Phase 1, pre- vs. post-training PI will be compared with a paired Wilcoxon signed-rank test. For Phase 2, PI across the three extinction trials will be compared with a Friedman test with post-hoc Wilcoxon tests. A significant PI decline in ATR-positive but not ATR-negative flies supports goal-directed learning; stable PI supports habitual responding.

References

- Aso Y, Sitaraman D, Ichinose T, Kaun KR, Vogt K, Belliart-Guerin G, et al. (2014) Mushroom body output neurons encode valence and guide memory-based action selection in *Drosophila*. *eLife* 3:e04580.
- Balleine BW, Dickinson A (1998) Goal-directed instrumental action: contingency and incentive learning and their cortical substrates. *Neuropharmacology* 37:407-419.
- Rajagopalan AE, Darshan R, Hibbard KL, Fitzgerald JE, Turner GC (2023) Reward expectations direct learning and drive operant matching in *Drosophila*. *Proc Natl Acad Sci USA*.
- Bouton ME (2024) Habit and persistence. *J Exp Anal Behav* 121:5-19.
- Brembs B (2009) Mushroom bodies regulate habit formation in *Drosophila*. *Curr Biol* 19:1351-1355.
- Dahanukar A, Lei YT, Kwon JY, Carlson JR (2007) Two Gr genes underlie sugar reception in *Drosophila*. *Neuron* 56:503-516.
- Dickinson A (1985) Actions and habits: the development of behavioral autonomy. *Philos Trans R Soc Lond B Biol Sci* 308:67-78.
- Felsenberg J, Jacob PF, Walker T, Barnstedt O, Edmondson-Stait AJ, Pleijzier MW, Otto N, Schlegel P, Sharifi N, Perisse E, Smith CS, Lauritzen JS, Costa M, Jefferis GSXE, Bhatt DL, Waddell S (2018) Integration of parallel opposing memories underlies memory extinction. *Cell* 175:709-722.
- Foucaud J, Burns JG, Mery F (2010) Use of spatial information and search strategies in a water maze analog in *Drosophila melanogaster*. *PLoS One* 5:e15231.
- Kahsai L, Zars T (2011) Learning and memory in *Drosophila*: behavior, genetics, and neural systems. *Int Rev Neurobiol* 99:139-167.
- Kim YC, Lee HG, Han KA (2007) D1 dopamine receptor dDA1 is required in the mushroom body neurons for aversive and appetitive learning in *Drosophila*. *J Neurosci* 27:7640-7647.
- Klapoetke NC, Murata Y, Kim SS, et al. (2014) Independent optical excitation of distinct neural populations. *Nat Methods* 11:338-346.
- Masek P, Scott K (2015) Gustatory learning and processing in the *Drosophila* mushroom bodies. *J Neurosci* 35:5950-5958.
- Pereira TD, Tabris N, Matsliah A, Turner DM, Li J, Ravindranath S, et al. (2022) SLEAP: A deep learning system for multi-animal pose tracking. *Nat Methods* 19:486-495.
- Scott K (2018) Gustatory processing in *Drosophila melanogaster*. *Annu Rev Entomol* 63:15-30.
- Tully T, Quinn WG (1985) Classical conditioning and retention in normal and mutant *Drosophila melanogaster*. *J Comp Physiol A* 157:263-277.
- Wiggin TD, Hsiao Y, Liu JB, Huber R, Griffith LC (2021) Rest is required to learn an appetitively-reinforced operant task in *Drosophila*. *Front Behav Neurosci* 15:681593.
- Wustmann G, Rein K, Wolf R, Heisenberg M (1996) A new paradigm for operant conditioning of *Drosophila melanogaster*. *J Comp Physiol A* 179:429-436.
- Yapici N, Zimmer M, Bhatt PK (2016) Closed-loop optogenetic activation of peripheral or central neurons modulates feeding in freely moving *Drosophila*. *eLife* 5:e45636.